

**Real-Time Anomaly Detection in IOMT Networks Using Stacking Model And A
Healthcare-Specific Dataset**

A MAIN PROJECT

**Submitted in partial fulfillment of the
Requirements for the award of the degree of
BACHELOR OF TECHNOLOGY**

In

CSE-CYBER SECURITY

By

BANDLA KRISHNA PRIYA (22F01A4606)

BONIGALA NANDINI (22F01A4610)

BATTULA YUVA DURGA (22F01A4607)

ASODI KEERTHI REDDY (22F01A4603)

SHAIK SABEEHA (22F01A4660)

Under the esteemed guidance

of

Dr. SREERAM INDRANEEL

PROFESSOR & HOD



DEPARTMENT OF CSE-CYBER SECURITY

**ST.ANN'S COLLEGE OF ENGINEERING & TECHNOLOGY, CHIRALA
(AUTONOMOUS)**

2022-2026

ST.ANN'S COLLEGE OF ENGINEERING & TECHNOLOGY, CHIRALA
(AUTONOMOUS)

DEPARTMENT OF CSE-CYBER SECURITY



CERTIFICATE

This is to certify that the Thesis entitled, **“REAL-TIME ANOMALY DETECTION IN IOMT NETWORKS USING STACKING MODEL AND A HEALTHCARE-SPECIFIC DATASET”** is being submitted by **BANDLA KRISHNA PRIYA, BONIGALA NANDINI, BATTULA YUVA DURGA, ASODI KEERTHI REDDY, SHAIK SABEEHA** bearing **Regd. No. 22F01A4606, 22F01A4610, 22F01A4607, 22F01A4603, 22F01A4660** in partial fulfillment of the requirement for the award of the degree of B.Tech in **CSE-CYBER SECURITY**, Jawaharlal Nehru Technological University Kakinada, is a record of Bonafide work carried out by them under my guidance and supervision. The results presented in this project have been verified and are found to be satisfactory.

Guide

Dr. SREERAM INDRANEEL
PROFESSOR & HOD

Head of the Department

Dr. SREERAM INDRANEEL
PROFESSOR & HOD

External Examiner

ACKNOWLEDGEMENTS

The successful completion of any task is not possible without proper suggestion, guidance and environment. Combination of these three factors acts like backbone to my project titled “**Real-Time Anomaly Detection in IOMT Networks Using Stacking Model And A Health Care Specific Dataset**”.

I express my sincere thanks to our guide **Dr. SREERAM INDRANEEL** and Head, Department of CSE- CYBER SECURITY, **Dr. SREERAM INDRANEEL** for his valuable suggestions during our course period, timely help, guidance and providing us with the most essential materials required for the completion of this work.

I am thankful to all teaching and non-teaching staff of the Department of CSE- CYBER SECURITY for the cooperation given for the successful completion of my project.

I would like to thank our principal, **Dr. K. JAGADEESH BABU** for providing support and simulating environment. I would like to express my gratitude to the **Management** of **St. Ann’s College of Engineering & Technology** for providing us a pleasant environment and excellent laboratory facilities.

B. Krishna Priya(22F01A4606)

B. Nandini(22F01A4610)

B. Yuva Durga(22F01A4607)

A. Keerthi Reddy(22F01A4603)

Sk. Sabeeha(22F01A4660)

ABSTRACT

The Internet of Medical Things (IoMT) integrates interconnected medical devices to enable real-time monitoring and personalized healthcare, but this connectivity also introduces serious cybersecurity risks such as data falsification, manipulation, and DOS attacks that threaten patient safety and service reliability. The objective of this study is to develop a real-time anomaly detection system capable of identifying malicious activities during IoMT data transmission. Manual monitoring approaches rely heavily on human observation and fixed rule based checking, which are slow, inconsistent, and unable to detect complex cyberattacks, making them unsuitable for high-speed IoMT environments. Previous system implementations using Random Forest, XGBoost, KNN, SVM and Artificial Neural Networks (ANN) provide enhanced detection capabilities but still struggle with processing efficiency, computational load, and adaptability to diverse medical attack scenarios. To address these limitations, this work proposes a machine learning based anomaly detection model using the Light Gradient Boosting Machine (LightGBM) algorithm. A new medical cybersecurity dataset inspired by UNSW-NB15 was developed, containing anomalous traffic and healthcare-specific attack types such as falsification and DoS patterns, providing a realistic and challenging evaluation environment. LightGBM was selected for its high efficiency, reduced computational overhead, and strong performance.

LIST OF TABLES

Table No	Name of the Table	Page No
Table 3.1	Datasets Used	13
Table 3.2	Preprocessing Summary	14
Table 3.3	Baseline Models	14
Table 3.4	Model Evaluation	16

LIST OF FIGURES

Figure No	Name of the Figure	Page No
Figure 4.1	Class Diagram	18
Figure 4.2	State Diagram	19
Figure 4.3	Sequence Diagram	20
Figure 4.4	Activity Diagram	20
Figure 4.5	Use Case Diagram	21
Figure 4.6	Component Diagram	22
Figure 4.7	Block Diagram	22
Figure 7.1	System Home Page	37
Figure 7.2	User Login Interface	37
Figure 7.3	Prediction Dashboard	38
Figure 7.4	Attack Prediction Page	38
Figure 7.5	Dataset Upload Window	39
Figure 7.6	Prediction Results Page	39

TABLE OF CONTENTS

CHAPTER	TITLE	PAGE NO
1	INTRODUCTION	1
2	LITERATURE SURVEY	4
	2.1 Machine learning - Based Anomaly Detection in Healthcare	5
	2.2 Ensemble Learning Approaches	6
	2.3 Deep Learning-Based Approaches	6
	2.4 Unsupervised and Semi-Supervised Techniques	7
	2.5 IoMT Security and Device-Level Anomaly Detection	7
	2.6 Dataset Challenges in IoMT	7
	2.7 Domain-Specific Applications	8
	2.8 Research Gap and Proposed Work	8
	2.9 Evaluation Metrics in Anomaly Detection Systems	9
	2.10 Real-Time Implementation Challenges in IoMT Systems	9
	2.11 Real-Time Implementation Challenges in IoMT Systems	10
3	PROPOSED WORK AND ANALYSIS	11
	3.1 Overview	12

	3.2 System Architecture	12
	3.3 Dataset Description and Integration	13
	3.4 Data Preprocessing and Feature Engineering	13
	3.5 Model Development	14
	3.6 Proposed Ensemble and LightGBM Model	15
	3.7 Model Evaluation	15
	3.8 Real-Time Implementation and Workflow	16
4	DESIGN	17
	4.1 Introduction to System Design	18
	4.2 UML Diagrams	18
	4.2.1 Class Diagram	18
	4.2.2 State Diagram	19
	4.2.3 Sequence Diagram	19
	4.2.4 Activity Diagram	20
	4.2.5 Use Case Diagram	21
	4.2.6 Component Diagram	21
	4.2.7 Block Diagram	22
5	IMPLEMENTATION	23

5.1	Software Requirements	24
5.2	Hardware Requirements	24
5.3	Implementation of Modules	25
5.3.1	Data Preparation	25
5.3.2	Feature Selection	26
5.3.3	Model Selection	27
5.4	Algorithms Used	28
5.5	Profitability Analysis	29
5.6	Coding	29
6	TESTING	33
7	RESULTS	36
8	CONCLUSION	40
9	FUTURE SCOPE	42
10	REFERENCES	44

INTRODUCTION

CHAPTER 1

INTRODUCTION

The Internet of Medical Things (IoMT) has revolutionized modern healthcare by seamlessly connecting medical devices, wearable sensors, and healthcare IT systems [1]. This interconnected ecosystem enables real-time patient monitoring, personalized treatment plans, and efficient care delivery, significantly enhancing healthcare quality and patient outcomes. Devices such as smart infusion pumps, wearable health monitors and connected diagnostic tools empower healthcare providers to make timely interventions and deliver tailored care strategies. Beyond clinical benefits, IoMT reduces costs, improves operational efficiency, and increases accessibility to medical services, holding immense potential to redefine healthcare systems globally [1], [2].

However, the integration of IoMT devices into healthcare networks introduces significant cybersecurity challenges. These devices often operate with limited computational resources and lack robust security features, making them vulnerable to cyberattacks such as data falsification, denial of service (DoS) attacks, and message tampering [3], [4]. Such threats can compromise the integrity and availability of critical healthcare systems, with severe consequences. For instance, a falsified message from an insulin pump could lead to incorrect dosages, or a DoS attack could disrupt vital monitoring systems during emergencies, endangering patient lives [5].

Traditional security measures, including cryptographic techniques like hashing and encryption, are essential but often impose computational demands that exceed the capabilities of resource-constrained IoMT devices [5], [6]. For example, ensuring message integrity through hashing can be computationally expensive for medical devices. Moreover, even encrypted data remains vulnerable to integrity and availability threats. To address these limitations, machine learning (ML)-based anomaly detection systems have emerged as a promising solution, enabling real-time analysis of incoming data to identify tampering, disruptions, or malicious activity.

To support this effort, we developed a new medical dataset for anomaly detection that reflects real-world healthcare scenarios and addresses the limitations of existing datasets. By incorporating features and attacks inspired by the UNSW-NB15 dataset [7] and introducing attacks that significantly impact healthcare systems, such as falsification and DoS attacks, this dataset provides a comprehensive foundation for evaluating Intrusion Detection Systems (IDS) solutions. It

includes diverse normal and anomalous data, balancing complexity and practicality for IoMT security research. Using this dataset, we evaluated seven ML algorithms, XGBoost [31], Random Forest [30], ANN, Support Vector Machines (SVM) [28], K-Nearest Neighbors (KNN) [27], Logistic Regression (LR) [29], and Isolation Forest (IF) [29], and selected XGBoost, Random Forest, and ANN as base learners for their strong accuracy and efficiency. These algorithms were integrated into a novel stacking ensemble model, with XGBoost serving as the meta-learner, to achieve robust and efficient anomaly detection tailored to IoMT environments.

This research addresses the cybersecurity challenges of IoMT, particularly data integrity and availability, through a real-time anomaly prediction model. The model dynamically adapts to incoming messages and newly detected anomalies or attack types. To overcome the lack of specialized datasets for healthcare systems, we developed a new medical dataset combining medical data with IoT network attacks (e.g., Fuzzers, Shellcode, Worms, Exploit) and healthcare-relevant attacks (e.g., Data Falsification and DoS). Additionally, we proposed a stacking ensemble model that demonstrates superior accuracy and efficiency, offering a reliable solution for real-time anomaly detection in healthcare environments

LITERATURE SURVEY

CHAPTER 2

LITERATURE SURVEY

The increasing use of machine learning for anomaly detection in Internet of Medical Things (IoMT) and Internet of Things (IoT) networks has led to diverse approaches, datasets, and methodologies. These systems generate large volumes of heterogeneous and real-time data, making traditional rule-based approaches insufficient. This section reviews key contributions in anomaly detection techniques, ensemble learning frameworks, and their applications in healthcare and IoT security. A comparison of prior works is provided in Appendix.

2.1 Machine Learning-Based Anomaly Detection in Healthcare

Machine learning techniques have been widely applied in healthcare systems to detect anomalies in physiological signals and patient monitoring data.

In [8], five machine learning algorithms were evaluated on the MIT-BIH dataset for heart rate anomaly detection. The study defined anomalies as values outside the fixed range of 60–100 bpm. Although the approach achieved good performance, it lacked adaptability to real-world conditions where normal heart rates vary across individuals. Local Outlier Factor (LOF) and Random Forest showed the best results, demonstrating the effectiveness of both density-based and ensemble methods.

In [9], the issue of missing labels was addressed using Generative Adversarial Networks (GANs) to generate synthetic anomaly labels. Classification was performed using Logistic Regression and XGBoost, while SHAP analysis was used to identify important features, improving interpretability in healthcare applications.

In [10], machine learning techniques such as Logistic Regression and Artificial Neural Networks (ANN) were applied for anomaly detection in smart healthcare systems. The results showed that ANN outperformed Logistic Regression, highlighting the advantage of nonlinear models in capturing complex data patterns

2.2 Ensemble Learning Approaches

Ensemble learning techniques combine multiple models to improve prediction accuracy and robustness, making them highly effective in IoMT environments.

In [\[12\]](#) ensemble learning methods such as Bagging, Boosting, and Stacking were evaluated for intrusion detection in IoMT systems using a healthcare dataset [\[13\]](#). While ensemble methods improved detection performance, the dataset's limited size and lack of diverse attack scenarios restricted comprehensive evaluation.

In [\[15\]](#), a hybrid ensemble model was proposed for detecting both known and zero-day Distributed Denial of Service (DDoS) attacks. The model achieved an accuracy of 99.1% on benchmark datasets, demonstrating the effectiveness of ensemble techniques in handling evolving threats.

In [\[20\]](#), a benchmarking study of multiple machine learning algorithms identified k-Nearest Neighbors, Decision Trees, and Naive Bayes as top-performing models for intrusion detection.

In [\[26\]](#), an explainable ensemble learning framework was proposed for detecting anomalies in blockchain transactions, improving classification performance through data balancing and feature importance analysis.

2.3 Deep Learning-Based Approaches

Deep learning models have gained popularity due to their ability to automatically extract meaningful features from high-dimensional data.

In [\[14\]](#), a Convolutional Neural Network (CNN)-based model was developed for anomaly detection in IoT networks, achieving high accuracy on standard datasets.

In [\[17\]](#), a deep autoencoder-based framework was introduced for detecting botnet attacks in IoT devices. Although effective, the study did not provide explicit accuracy metrics.

In [\[22\]](#), Stacked Autoencoders (SAE) were used for early fault detection in industrial systems, demonstrating strong performance in time-series anomaly detection.

In [\[18\]](#), a semi-supervised deep learning model was proposed for DDoS attack detection, though it was limited to specific attack types.

2.4 Unsupervised and Semi-Supervised Techniques

Unsupervised and semi-supervised approaches are useful in IoMT systems where labeled data is limited.

In [16], a semi-supervised K-means algorithm was proposed for DDoS detection. While the method showed potential, it lacked detailed accuracy evaluation.

In [23], dimensionality reduction using Principal Component Analysis (PCA) combined with One-Class Support Vector Machines (OC-SVM) was explored for anomaly-based intrusion detection.

In [24], a machine learning-based anomaly detection method was proposed for Wireless Sensor Networks, focusing on reducing computational complexity.

2.5 IoMT Security and Device-Level Anomaly Detection

In addition to data anomalies, detecting abnormal device behavior is critical in IoMT systems.

In [11], an anomaly detection model was proposed for protecting medical IoT control services against external attacks, focusing on securing device communication.

In [19], a machine learning-based DDoS detection system for IoT devices was presented, achieving strong performance under controlled conditions.

In [25], multiple machine learning algorithms were evaluated for IoT attack detection, with improved performance achieved through feature engineering.

2.6 Dataset Challenges in IoMT

The effectiveness of anomaly detection models depends heavily on the quality and diversity of datasets.

Commonly used datasets include NSL-KDD, UNSW-NB15, CICIDS2017, and BoT-IoT. While these datasets are widely accepted, they lack healthcare-specific attributes required for IoMT applications.

In [13], a dataset designed for healthcare monitoring systems was introduced. However, its limited size and restricted attack diversity reduce its effectiveness for large-scale evaluation.

Additionally, some studies rely on synthetic datasets, which may not accurately represent real-world scenarios.

2.7 Domain-Specific Applications

Anomaly detection techniques are applied across various domains, including healthcare, industrial systems, and financial applications.

In [21], deep learning techniques for time-series anomaly detection were reviewed, providing insights into model selection and performance.

In [22], deep learning models were successfully applied for industrial fault detection.

In [26], machine learning and explainable AI techniques were used for anomaly detection in financial transaction systems.

These studies demonstrate the adaptability of anomaly detection techniques across different domains.

2.8 Research Gap and Proposed Work

Despite significant progress, several limitations remain in existing research:

- Lack of comprehensive healthcare-specific datasets
- Limited real-time implementation of anomaly detection systems
- Over-reliance on benchmark or synthetic datasets
- Limited integration of medical and network anomaly detection

To address these gaps, this research proposes a novel approach that combines medical data with attack patterns inspired by benchmark datasets. The dataset is enriched with healthcare-specific threats to improve realism.

A stacking ensemble model integrating XGBoost, Random Forest, and Artificial Neural Networks (ANN) is proposed. The model is evaluated in real-time scenarios, demonstrating improved accuracy, robustness, and efficiency for IoMT healthcare applications.

2.9 Evaluation Metrics in Anomaly Detection Systems

Evaluation metrics play a crucial role in assessing the performance of anomaly detection models in IoMT and IoT environments. Since these systems often deal with imbalanced datasets, relying solely on accuracy may lead to misleading conclusions.

In [20], multiple performance metrics such as accuracy, precision, recall, and F1-score were used to evaluate machine learning models on intrusion detection datasets. The study emphasized that precision and recall are more important in anomaly detection, as they measure the ability to correctly identify attacks and minimize false alarms.

In [23], the performance of anomaly-based intrusion detection systems was analyzed using classification metrics along with detection rate and false positive rate. The study highlighted that a high detection rate with a low false positive rate is essential for practical deployment.

In [15], evaluation of ensemble models for DDoS detection included metrics such as confusion matrix analysis and receiver operating characteristic (ROC) curves. These metrics provided a comprehensive understanding of model performance across different attack scenarios.

Additionally, real-time IoMT systems require evaluation metrics that consider latency and computational efficiency. Models with high accuracy but high computational cost may not be suitable for deployment in resource-constrained environments.

2.10 Feature Engineering and Data Preprocessing Techniques

Feature engineering and data preprocessing significantly impact the performance of anomaly detection systems in IoMT networks.

In [25], feature engineering techniques were applied to improve the detection accuracy of machine learning models on IoT datasets. The introduction of new features enhanced the ability of models to distinguish between normal and malicious traffic.

In [16], hybrid feature selection methods were used in conjunction with clustering algorithms to improve anomaly detection performance. Feature selection helped reduce dimensionality and improved computational efficiency.

In [26], data balancing techniques were introduced to address class imbalance in anomaly detection tasks. The proposed approach improved classification performance and reduced bias toward majority classes.

In [23], dimensionality reduction using Principal Component Analysis (PCA) was used to remove redundant features and improve model scalability. This approach is particularly useful in IoMT systems where high-dimensional data is common.

Effective preprocessing techniques such as normalization, noise removal, and feature scaling are essential for improving model performance and ensuring reliable anomaly detection.

2.11 Real-Time Implementation Challenges in IoMT Systems

Real-time anomaly detection in IoMT systems presents several challenges due to the dynamic and resource-constrained nature of healthcare environments.

In [11], anomaly detection models were designed to protect medical IoT control services. However, real-time implementation requires handling continuous data streams and ensuring low latency.

In [14], deep learning models demonstrated high accuracy but often required significant computational resources, making them less suitable for real-time deployment on edge devices.

In [18], the proposed model addressed DDoS detection in IoT environments, but its applicability was limited by the need for specific infrastructure such as SDN-based architectures.

In [19], machine learning-based detection systems showed strong performance in controlled environments, but real-world deployment may introduce challenges such as network variability and unseen attack patterns.

Addressing these challenges is essential for developing practical and deployable anomaly detection systems in healthcare environments.

PROPOSED METHODOLOGY

CHAPTER 3

PROPOSED WORK AND ANALYSIS

3.1 Overview

This chapter describes the proposed methodology for anomaly detection in Internet of Medical Things (IoMT) environments. The system is designed to detect cyber threats in healthcare networks by integrating machine learning techniques with a hybrid dataset combining network intrusion and healthcare data.

The methodology follows a structured pipeline consisting of data collection, preprocessing, feature engineering, model development, and evaluation. The proposed system uses a stacking ensemble approach along with LightGBM to improve detection accuracy and reduce false positives.

3.2 System Architecture

The system is designed using a layered architecture that ensures efficient data processing and accurate prediction. The architecture includes the following components:

- Data Collection Layer
- Preprocessing Layer
- Feature Engineering Layer
- Model Training Layer
- Ensemble Learning Layer
- Evaluation Layer
- Real-Time Prediction Layer

These layers work sequentially to transform raw IoMT data into meaningful predictions for anomaly detection.

3.3 Dataset Description and Integration

The proposed system utilizes a hybrid dataset formed by combining network intrusion data with healthcare data to simulate real-world IoMT scenarios.

Table 3.1 Datasets Used

Dataset	Description	Purpose
UNSW-NB15	Network traffic dataset	Detect cyber attacks
Diabetes	Healthcare dataset	Patient health data

The UNSW dataset includes features such as packet flow, protocols, and attack categories, while the healthcare dataset includes patient-related attributes.

Dataset Integration

Both datasets are merged to form a unified dataset that captures interactions between medical data and network traffic. This improves anomaly detection by providing contextual understanding of IoMT environments.

3.4 Data Preprocessing and Feature Engineering

Data preprocessing ensures data quality and consistency before model training.

Preprocessing Steps

- Removal of duplicate values
- Handling missing values using mean imputation
- Encoding categorical variables
- Feature scaling using standardization

Feature Engineering

- Selection of important features
- Reduction of irrelevant attributes
- Optional dimensionality reduction techniques

Table 3.2 Preprocessing Summary

Step	Technique Used	Purpose
Missing Values	Mean Imputation	Avoid data loss
Duplicates	Removal	Reduce noise
Encoding	Label Encoding	Convert categorical data
Scaling	Standardization	Normalize features

3.5 Model Development

To build an effective anomaly detection system, multiple machine learning models are implemented and compared.

Table 3.3 Baseline Models

Algorithm	Purpose
Logistic Regression	Baseline classification
SVM	High-dimensional data handling
Random Forest	Ensemble learning
KNN	Distance-based learning
Isolation Forest	Anomaly detection
ANN	Complex pattern learning
XGBoost	Boosting performance

These models help in understanding performance differences and selecting the best approach.

3.6 Proposed Ensemble and LightGBM Model

The proposed system combines multiple models using a stacking ensemble approach and further enhances performance using LightGBM.

Stacking Ensemble

- Combines predictions of multiple base learners
- Improves accuracy and robustness
- Reduces overfitting

LightGBM Model

- Fast and efficient gradient boosting algorithm
- Handles large datasets effectively
- Provides better accuracy compared to traditional models

This hybrid approach ensures high performance in detecting anomalies in IoMT systems.

3.7 Model Evaluation

The trained models are evaluated using standard performance metrics to ensure reliability.

Table 3.4 Model Evaluation

Metric	Description
Accuracy	Overall correctness
Precision	Correct positive predictions
Recall	Ability to detect attacks
F1-Score	Balance of precision and recall

The dataset is split into:

- 80% training data
- 20% testing data

This ensures unbiased evaluation and proper generalization.

3.8 Real-Time Implementation and Work flow

The final model is deployed for real-time anomaly detection in IoMT environments.

Work flow Steps

1. Data collection from IoMT devices
2. Data preprocessing and cleaning
3. Dataset integration
4. Feature engineering
5. Model training and stacking
6. LightGBM optimization
7. Model evaluation
8. Real-time prediction

DESIGN

CHAPTER 4

DESIGN

4.1 Introduction to System Design

System design is a crucial phase in the development of any software system, as it defines the overall architecture, components, data flow, and interactions within the system. In the context of the proposed anomaly detection system for Internet of Medical Things (IoMT), system design focuses on creating an efficient, scalable, and secure framework capable of processing large volumes of healthcare and network data.

4.2 UML DIAGRAMS

4.2.1 Class Diagram

A class diagram represents the **static structure of a system** by illustrating its classes, attributes, methods, and relationships. It provides a blueprint of how data and functionality are encapsulated within the system and how different components are logically connected.

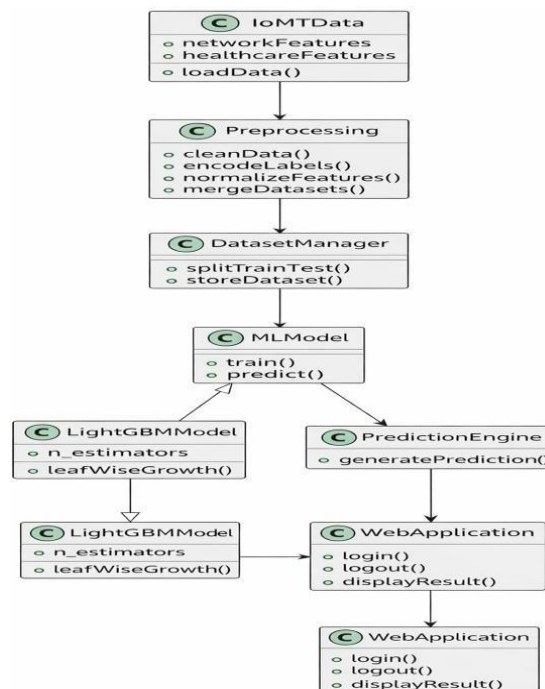


Figure 4.1: Class Diagram

4.2.2 State Diagram

A state diagram represents the different states of a system and the transitions between those states based on events or conditions. It helps in understanding how the system behaves dynamically over time.

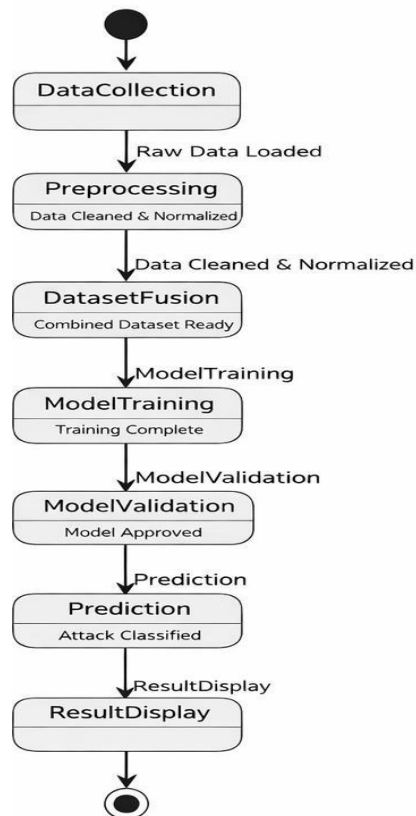


Figure 4.2: State Diagram

4.2.3 Sequence Diagram

A sequence diagram depicts the **time-ordered interaction between system components**, showing how objects communicate through messages to perform a specific operation. It emphasizes the dynamic behavior of the system and clarifies the execution flow from request initiation to final response. In IoMT anomaly detection systems, sequence diagrams are particularly useful for illustrating how user requests are processed through preprocessing, machine learning inference, and result delivery.

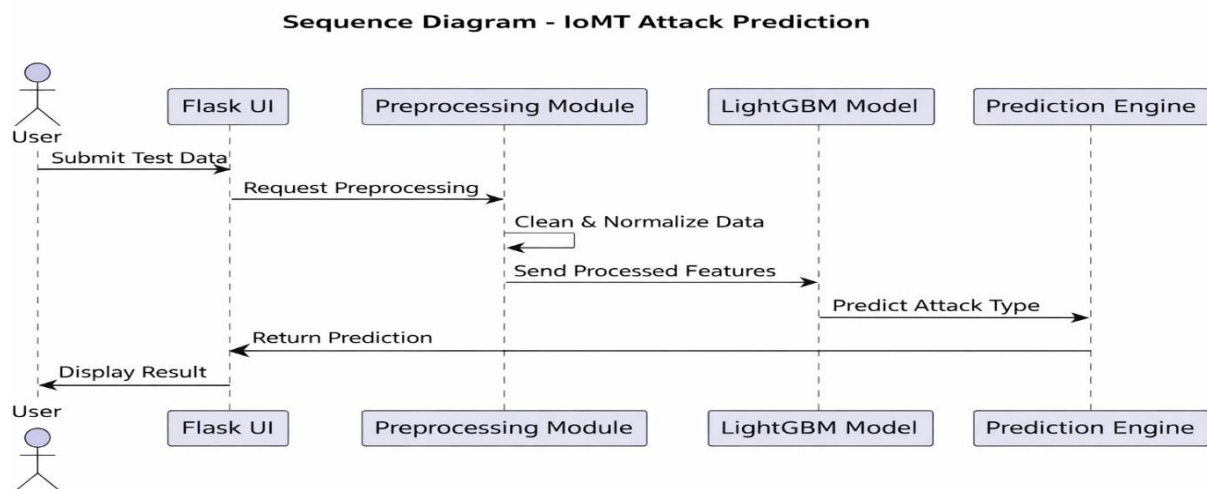


Figure 4.3: Sequence Diagram

4.2.4 Activity Diagram

An activity diagram represents the **workflow of activities and decision points** within a system. It focuses on the sequence of operations, including branching and parallel processing, rather than object interaction.

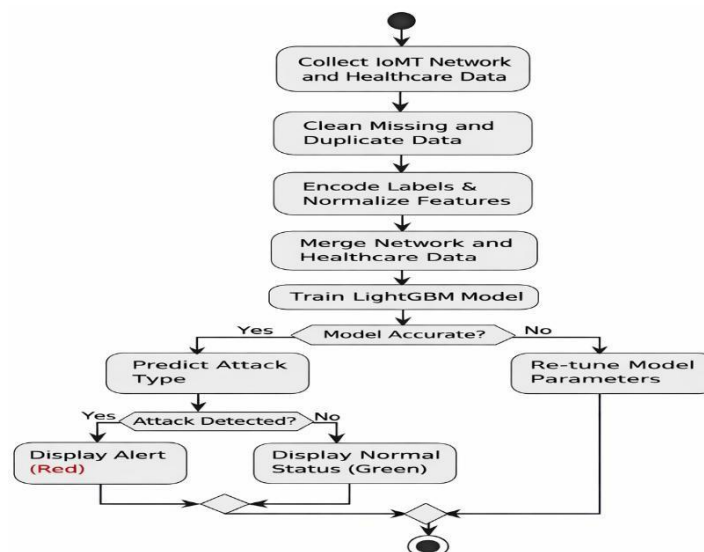


Figure 4.4: Activity Diagram

4.2.5 Use Case Diagram

A use case diagram describes **what the system does from the user's perspective**. It identifies system functionalities (use cases) and shows how different actors interact with those functionalities. Use case diagrams are helpful for capturing **functional requirements** and defining system boundaries.

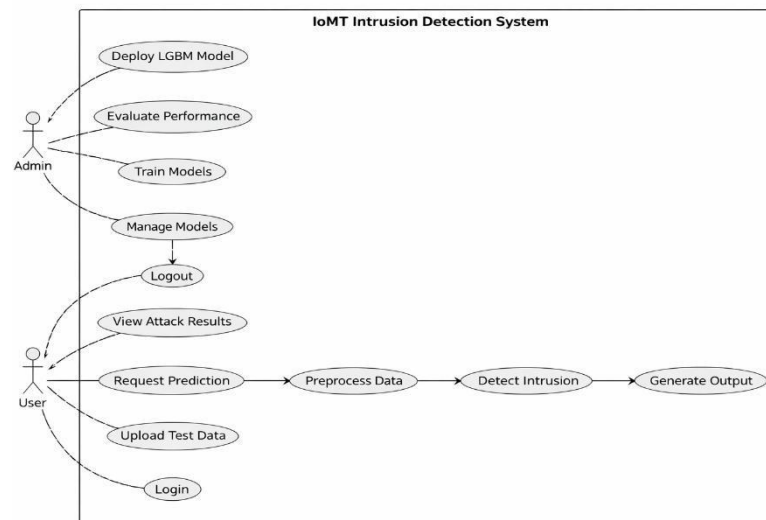


Figure 4.5: Use Case Diagram

4.2.6 Component Diagram

A component diagram illustrates the **structural organization of a system**, showing how software components interact through interfaces. It focuses on **modularity, dependencies, and high-level architecture**, making it useful for understanding system design and deployment structure.

The **main purpose of a Component Diagram** is to represent the **overall architecture of a software system** by dividing it into smaller, manageable modules called components. Each component performs a specific function and interacts with other components through well-defined interfaces. By illustrating these relationships, the diagram helps developers understand how different parts of the system communicate and depend on each other. Component diagrams are especially useful in large and complex systems because they simplify the structure and make it easier to design, develop, and maintain the application. They also support better planning, system integration, and reuse of software components in different projects.

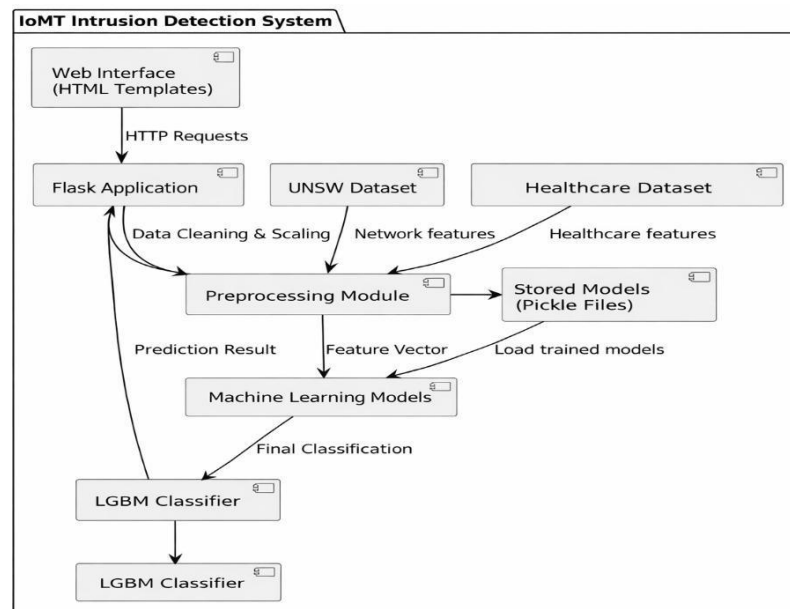


Figure 4.6 Component Diagram

4.2.7 Block Diagram

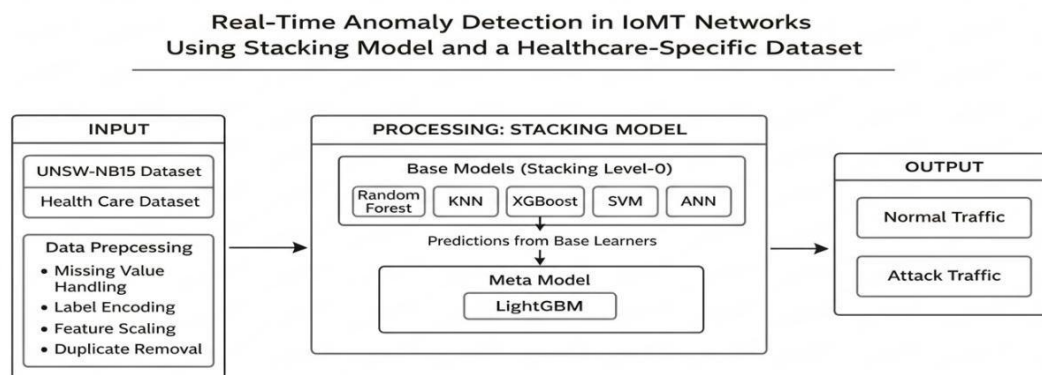


Figure 4.7: Block Diagram

IMPLEMENTATION

CHAPTER 5

IMPLEMENTATION

5.1 Software Requirements

The implementation of the proposed system for real-time anomaly detection in IoMT networks requires a robust software environment capable of handling large-scale healthcare data and machine learning operations.

- Operating System: Windows 10/11 or Ubuntu Linux
- Programming Language: Python (due to its strong support for machine learning and data analysis)
- Development Tools: Jupyter Notebook, VS Code
- Libraries Used:
 - NumPy – numerical computations
 - Pandas – data preprocessing and manipulation
 - Matplotlib & Seaborn – data visualization
 - Scikit-learn – machine learning models and evaluation
 - LightGBM – high-performance gradient boosting framework
- Version Control: Git for code management
- Documentation Tools: MS Word, PowerPoint

These tools collectively enable efficient development, training, and evaluation of the anomaly detection system.

5.2 Hardware Requirements

The system is designed to operate efficiently on standard computing devices while also supporting scalability.

- **Processor:** Intel Core i5 or higher

- **RAM:** Minimum 8 GB (16 GB recommended)
- **Storage:** 256 GB SSD or higher
- **GPU (Optional):** NVIDIA GPU for faster training
- **Internet:** Required for dataset access and updates

These specifications ensure smooth handling of IoMT datasets and model training.

5.3 Implementation of Modules

The proposed system is divided into modular components to ensure flexibility, scalability, and ease of maintenance. Each module performs a specific task in the anomaly detection pipeline.

5.3.1 Data Preparation

Data preparation is a crucial step in building an effective anomaly detection model.

- The dataset used in this project is a **healthcare/IoMT network dataset**, containing multiple records of network traffic and device behavior.
- Each record includes features such as:
 - Packet size
 - Transmission rate
 - Device activity patterns
 - Protocol information

Steps Involved:

1. Data Cleaning

- Missing values are handled using mean or median imputation.
- Duplicate records are removed.

2. Data Transformation

- Numerical features are normalized using Min-Max scaling.

- Standardization is applied where necessary.

3. Handling Outliers

- Extreme values are analyzed and treated carefully, as anomalies are important in this project.

4. Encoding

- Categorical data is converted into numerical form using Label Encoding.

5. Dataset Description

- Total records (Health care Dataset):768
- Features:9
- Target variable: Normal vs Anomaly

Proper preprocessing ensures that the dataset is clean and suitable for training accurate models.

5.3.2 Feature Selection

Feature selection helps in identifying the most relevant attributes that contribute to anomaly detection.

Techniques Used:

- **Correlation Matrix**
 - Removes highly correlated and redundant features.
- **Feature Importance (Tree-Based Models)**
 - Random Forest and LightGBM are used to rank feature importance.
- **Dimensionality Reduction**
 - Unnecessary features are eliminated to reduce computational complexity.

Benefits:

- Improves model accuracy

- Reduces overfitting
- Decreases training time

The selected features significantly enhance the performance of the final model.

5.3.3 Model Selection

Different machine learning models are evaluated to determine the best-performing algorithm.

Models Considered:

- Logistic Regression
- Decision Tree
- Random Forest
- Support Vector Machine (SVM)
- LightGBM (Proposed Model)

Evaluation Metrics:

- Accuracy
- Precision
- Recall
- F1-Score

Process:

- Dataset is split into training and testing sets
- Each model is trained and evaluated
- Performance comparison is conducted

The results indicate that **LightGBM outperforms other models**, making it the primary algorithm in this project.

5.4 Algorithms Used

The system employs multiple machine learning algorithms, with LightGBM as the core component.

1. Logistic Regression

Used for baseline comparison in binary classification tasks.

2. Decision Tree

Provides a simple and interpretable structure for decision-making.

3. Random Forest

An ensemble method that improves prediction accuracy by combining multiple trees.

4. Support Vector Machine (SVM)

Effective in handling high-dimensional data and classification tasks.

5. Light Gradient Boosting Machine (LightGBM) – Proposed Algorithm

LightGBM is the primary algorithm used in this project due to its efficiency and high performance.

Key Features:

- Leaf-wise tree growth for better accuracy
- Faster training using histogram-based learning
- Handles large datasets efficiently

Parameters Used:

- `learning_rate = 0.1`
- `n_estimators = 100`
- `max_depth = -1`
- `boosting_type = 'gbdt'`

Advantages in This Project:

- Detects anomalies with high precision
- Reduces training time significantly
- Works well with complex IoMT data

6. Stacking Model

Stacking combines multiple models to improve prediction performance.

- Base Models: Random Forest, SVM, LightGBM
- Meta Model: Logistic Regression

The stacking approach enhances accuracy by leveraging strengths of different models.

5.5 Profitability Analysis

The proposed system offers significant practical and economic benefits:

- **Reduced Manual Monitoring:** Automates anomaly detection
- **Improved Accuracy:** Detects threats more effectively
- **Time Efficiency:** Provides faster predictions
- **Scalability:** Handles large IoMT networks
- **Cost Reduction:** Minimizes operational expenses

This makes the system highly suitable for real-world healthcare environments.

5.6 Coding

The implementation is carried out using Python in a structured manner.

Steps Involved:

1. Importing libraries
2. Loading dataset

3. Data preprocessing
4. Splitting dataset
5. Training models
6. Prediction
7. Evaluation

SAMPLE CODE

IMPORT LIBRARIES

```
import pandas as pd
import numpy as np
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier, StackingClassifier
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score
from lightgbm import LGBMClassifier
```

LOAD DATASET

```
dataset = pd.read_csv("UNSW_NB15.csv")
health_dataset = pd.read_csv("healthcare.csv")
```

DATA PREPROCESSING

Encode categorical data

```
le = LabelEncoder()
dataset['attack_cat'] = le.fit_transform(dataset['attack_cat'])
```

Handle missing values

```
dataset.fillna(dataset.mean(), inplace=True)
health_dataset.fillna(health_dataset.mean(), inplace=True)
```

Split features and target

```
X = dataset.drop('attack_cat', axis=1)
y = dataset['attack_cat']
```

Normalize data

```
scaler = StandardScaler()
X = scaler.fit_transform(X)
```

MERGE DATASETS

```
health_X = StandardScaler().fit_transform(health_dataset)
X_combined = np.hstack((X, health_X))
```

Train-test split

```
X_train, X_test, y_train, y_test = train_test_split(
    X_combined, y, test_size=0.2, random_state=42
)
```

MODEL TRAINING (Base Models)

```
rf = RandomForestClassifier()
lgbm = LGBMClassifier(n_estimators=200)
```

STACKING MODEL

```
stacking_model = StackingClassifier(  
    estimators=[  
        ('rf', rf),  
        ('lgbm', lgbm)  
    ],  
    final_estimator=LogisticRegression(max_iter=1000)  
)
```

Train stacking model

```
stacking_model.fit(X_train, y_train)
```

PREDICTION

```
y_pred = stacking_model.predict(X_test)
```

PERFORMANCE METRICS

```
print("Accuracy :", accuracy_score(y_test, y_pred))  
print("Precision:", precision_score(y_test, y_pred, average='macro'))  
print("Recall  :", recall_score(y_test, y_pred, average='macro'))  
print("F1 Score :", f1_score(y_test, y_pred, average='macro'))
```

TESTING

CHAPTER 6

TESTING

Test Case ID	Test Scenario	Test Steps	Expected Result	Actual Result	Status
TC_01	Dataset Loading	Load UNSW-NB15 and Healthcare datasets	Datasets should be loaded with all required attributes	Both datasets loaded with complete feature set	Pass
TC_02	Data Preprocessing	Perform missing value handling and encoding	Cleaned dataset with no null values and encoded features	Preprocessing completed with no missing values	Pass
TC_03	Feature Scaling	Apply normalization using StandardScaler	All feature values scaled within standard range	Features normalized successfully	Pass
TC_04	Dataset Integration	Merge UNSW and healthcare datasets	Combined dataset formed with aligned features	Dataset merged without inconsistencies	Pass
TC_05	Data Splitting	Perform 80:20 train-test split	Training and testing datasets created properly	Data split executed without data loss	Pass

TC_06	Base Model Training	Train Random Forest models	Models should converge without errors	Models trained with stable performance	Pass
TC_07	LightGBM Model Training	Train LightGBM on combined dataset	Model should achieve high accuracy and fast convergence	LightGBM trained with improved accuracy	Pass
TC_08	Ensemble Model (Stacking)	Apply Stacking (Ensemble Learning) for model combination using a meta-classifier	Ensemble should enhance prediction performance	Improved prediction performance achieved using stacking ensemble	Pass
TC_09	Anomaly Prediction	Predict anomalies on test dataset	System should classify Normal and Attack categories correctly	Predictions generated with correct classification	Pass
TC_10	Performance Evaluation	Compute Accuracy, Precision, Recall, F1-score	Metrics should reflect model performance accurately	Metrics computed successfully	Pass

RESULTS

CHAPTER 7

RESULTS

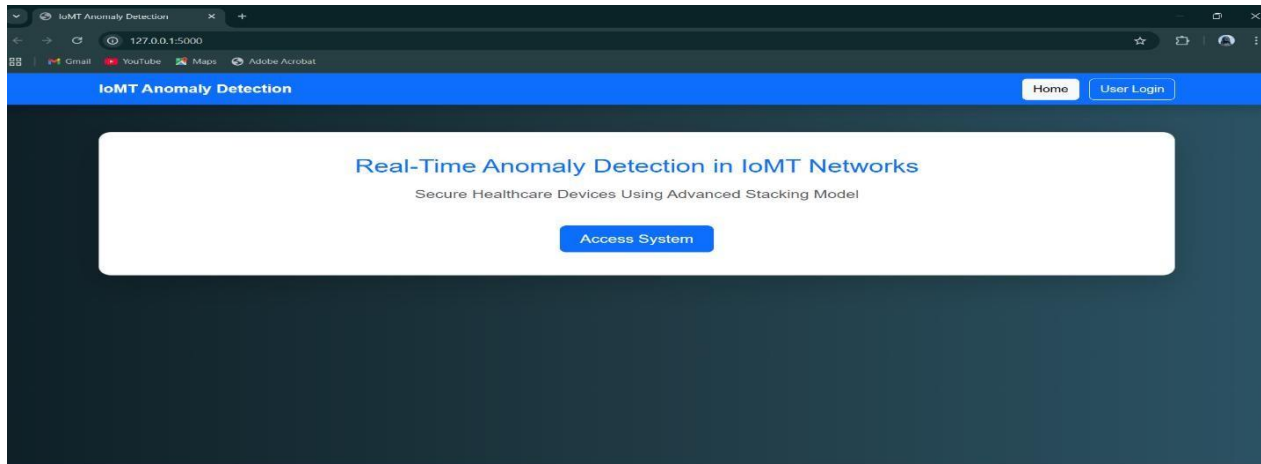


Figure7.1: System Home Page

Figure 7.1 presents a web-based **IoMT Anomaly Detection System** designed for real-time monitoring of healthcare device networks. It highlights a stacking model approach to enhance security and detect abnormal activities efficiently. The clean dashboard includes navigation options and an “Access System” button for user interaction.

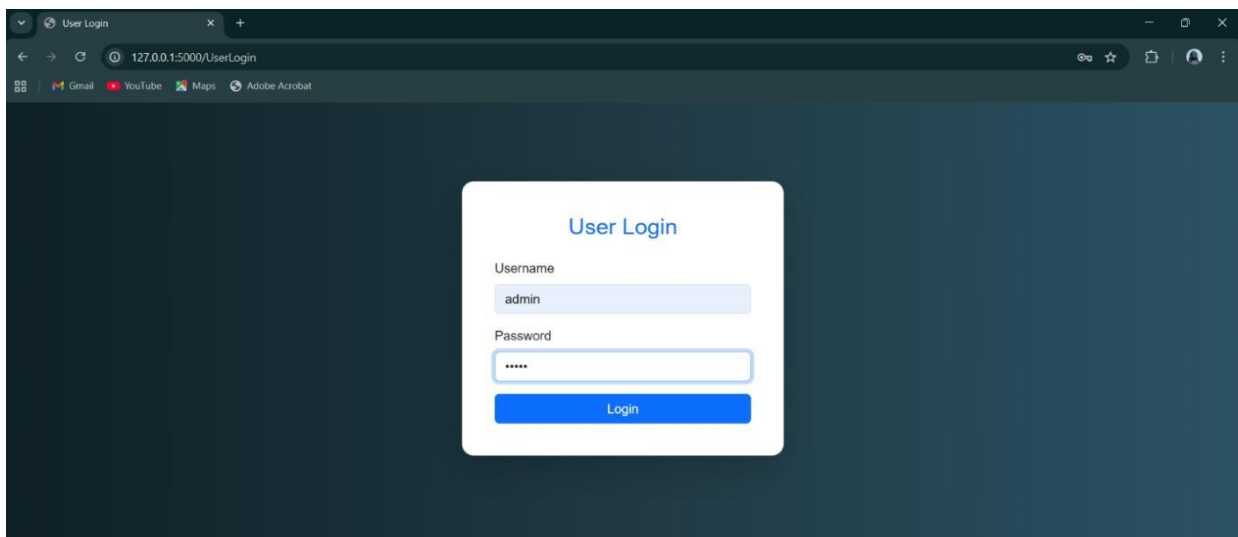


Figure7.2: User Login Interface

Figure 7.2 shows secure access to the IoMT Anomaly Detection System. It requires authorized credentials, including a username and password, to ensure that only authenticated users

can access network monitoring and analysis features. The simple and centered design enhances usability while maintaining system security.

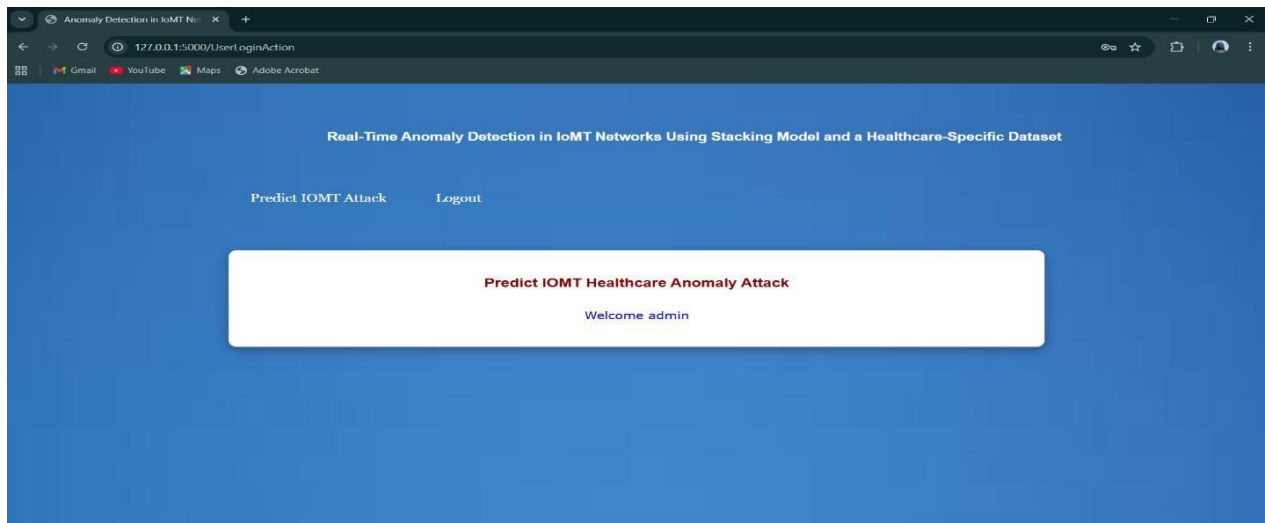


Figure 7.3: Prediction Dashboard

Figure 7.3 displays the main prediction interface of the IoMT anomaly detection system. It allows the admin to initiate healthcare attack prediction using the trained stacking model and provides navigation options such as attack prediction and logout for secure session management.

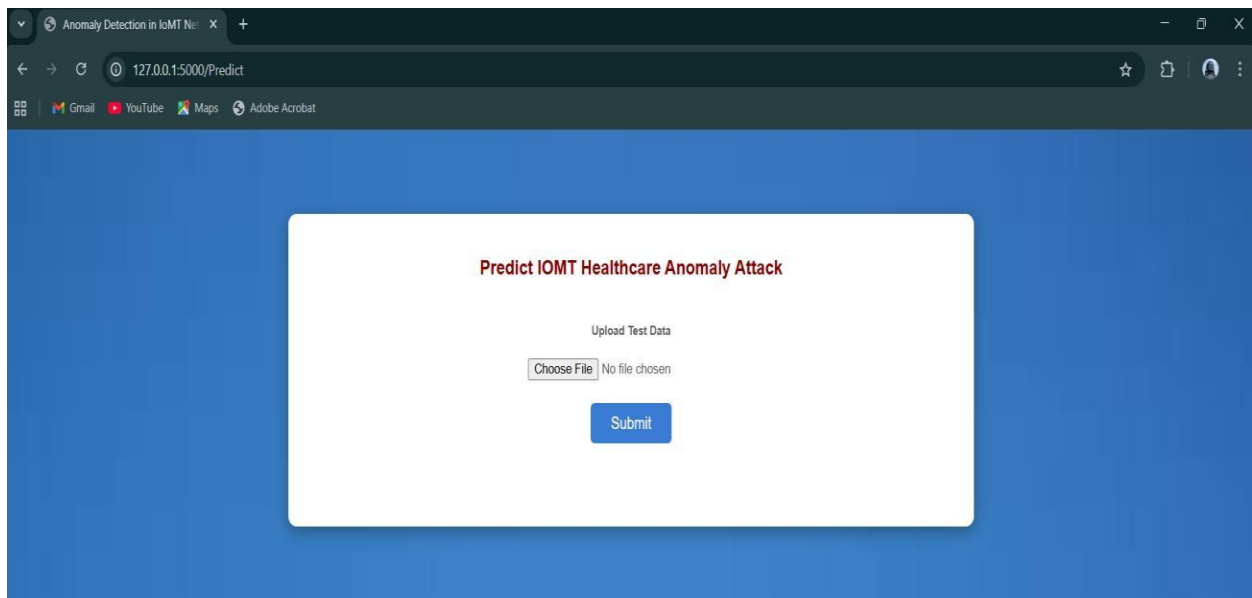


Figure 7.4: Attack Prediction Page

Figure 7.4 shows the user to upload IoMT healthcare test data for anomaly detection. After submitting the dataset, the trained stacking model analyzes the input and predicts whether the network traffic is normal or an attack.

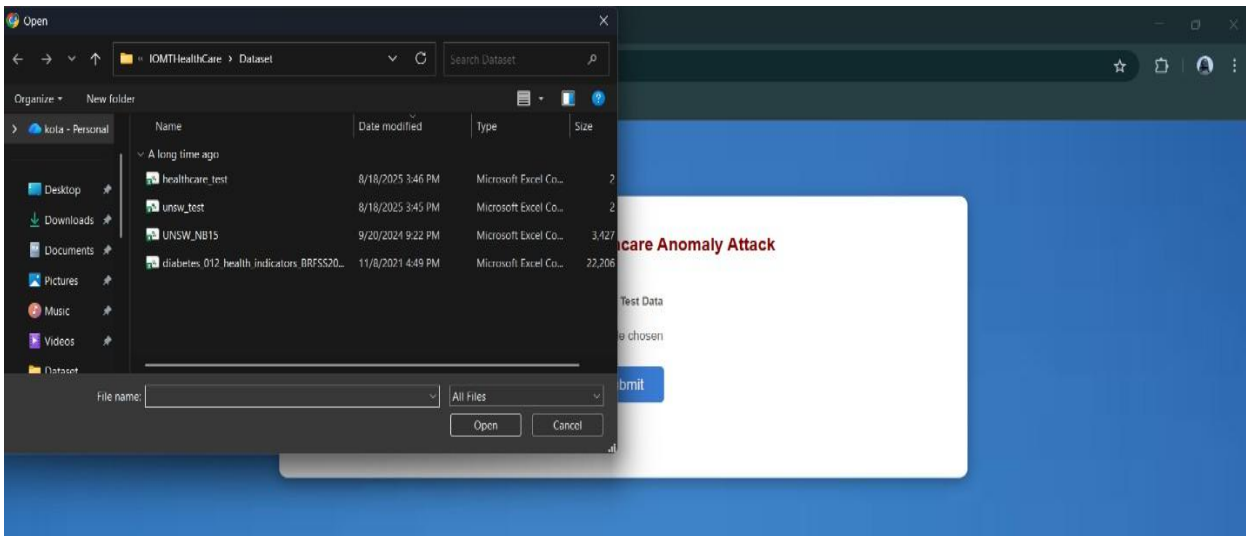


Figure 7.5: Dataset Upload Window

Figure 7.5 shows the user to select a healthcare test dataset from the local system. The chosen file is then uploaded to the IoMT anomaly detection platform for analysis and attack prediction using the trained model.

Test Data	Predicted Attack Type
[-0.33333333 -3. -0.33333333 0.01310153 -0.44962547 -2.65495395 -0.3506581 -0.31642315 -0.41071765 -0.39570649 0. -0.81649658 -0.5 0. 0.47664128 -0.65465367 0. -0.33333333 0.65465367 0.5 0. -0.33333333 0. 0. -0.77777778 -0.74478352 -0.55848937 -0.33333333 -1. -0.66208471 0.75 0.8046798]	Normal
[3. 0.33333333 3. -0.81666202 -0.94920932 -1.09321633 -0.60112817 -1.0195857 -0.60174912 -0.84537297 0. -0.81649658 -0.5 0. -1.02854171 1.52752523 0. -0.33333333 0.65465367 0.5 0. -0.33333333 0. 0. 0.33333333 0.78859432 -0.19425717 -0.33333333 1. 0.57932412 0.75 0.8046798]	Worms
[-0.33333333 0.33333333 -0.33333333 -0.3362726 0.88259814 0.46852129 0.02504701 0.73832068 -0.12417045 0.59355974 0. 1.22474487 -0.5 0. 0.22577745 -0.65465367 0. 3. -1.52752523 0.5 0. -0.33333333 0. 0. 1.44444444 0.13143239 -0.55848937 3. -1. 1.82073295 -1.75 -1.67125804]	Reconnaissance
[-0.33333333 0.33333333 -0.33333333 -0.3362726 0.88259814 0.46852129 0.02504701 0.73832068 -0.12417045 0.59355974 0. 1.22474487 -0.5 0. 0.22577745 -0.65465367 0. -0.33333333 -1.52752523 -2. 0. -0.33333333 0. 0. 0.33333333 0.35048636 -0.55848937 -0.33333333 1. -1.4896906 0.75 -1.67125804]	Fuzzers
[-0.33333333 0.33333333 -0.33333333 0.48257301 -0.94920932 0.46852129 -0.60112817 -1.0195857 -0.60174912 -0.93530626 0. -0.81649658 -0.5 0. -1.53026937 -0.65465367 0. -0.33333333 0.65465367 0.5 0. -0.33333333 0. 0. -0.77777778 -0.74478352 -0.55848937 -0.33333333 -1. 0.57932412 0.75 0.18569534]	Exploits
[-0.33333333 0.33333333 -0.33333333 -0.3362726 -0.44962547 0.46852129 -0.3506581 -0.31642315 -0.02865472 -0.39570649 0. 1.22474487 2. 0. -0.27595021 1.52752523 0. -0.33333333 0.65465367 0.5 0. -0.33333333 0. 0. 0.33333333 -0.74478352 -0.55848937 -0.33333333 1. 1.40693001 -0.5 0.8046798]	DoS
[-0.33333333 0.33333333 -0.33333333 2.80809454 -0.94920932 0.46852129 -0.60112817 -1.0195857 -0.60174912 -0.93530626 0. -0.81649658 -0.5 0. -0.77767788 -0.65465367 0. -0.33333333 0.65465367 0.5 0. -0.33333333 0. 0. 0.33333333 0.74478352 -0.19425717 -0.33333333 -1. -0.66208471 -1.75 0.18569534]	Exploits

Figure 7.6: Prediction Results Page

Figure 7.6 displays the uploaded test dataset along with the corresponding predicted attack types generated by the stacking model. It classifies each network instance as Normal or attack categories.

CONCLUSION

CHAPTER 8

CONCLUSION

This research successfully developed an effective anomaly detection framework for Internet of Medical Things (IoMT) environments by integrating network intrusion data with healthcare indicators and employing advanced machine learning techniques. Experimental results demonstrate that while individual models such as Artificial Neural Networks (ANN) achieved reasonable performance, their detection capability was limited when applied independently to complex and heterogeneous IoMT data. Specifically, ANN achieved an accuracy of approximately 82.06% on the UNSW dataset and 85.99% on the combined healthcare dataset, indicating moderate learning capability but insufficient generalization for high-risk medical cyber environments.

The proposed Meta Model based on LightGBM significantly outperformed standalone models by effectively learning hierarchical feature interactions from the fused dataset. With an accuracy of 98.31%, precision of 98.14%, recall of 96.13%, and F1-score of 97.08%, the LightGBM-based Meta Model demonstrated superior classification reliability and robustness. These results confirm that integrating healthcare context with network traffic data greatly enhances anomaly detection accuracy and reduces false positives, which is critical in safety-critical IoMT systems.

Furthermore, the high recall achieved by the proposed approach indicates strong capability in identifying diverse attack categories, minimizing the risk of undetected intrusions. The improvement in F1-score highlights a balanced trade-off between precision and recall, ensuring dependable decision-making. Overall, this research validates the effectiveness of LightGBM as a Meta Model for real-time IoMT cybersecurity, providing a scalable and accurate solution suitable for practical healthcare deployments.

FUTURE SCOPE

CHAPTER 9

FUTURE SCOPE

Although the proposed LightGBM-based Meta Model demonstrates excellent performance, several avenues exist for further enhancement. Future work can explore real-time streaming data integration using platforms such as Apache Kafka or MQTT to enable continuous intrusion monitoring in live IoMT environments. This would allow the system to adapt dynamically to evolving attack patterns and support proactive defense mechanisms.

Another promising direction involves incorporating advanced deep learning architectures such as Graph Neural Networks (GNNs) or Transformer-based models to capture temporal and relational dependencies among IoMT devices. Additionally, implementing automated hyperparameter optimization and adaptive learning strategies could further improve detection accuracy while reducing manual tuning effort. Future research may also integrate explainable AI techniques to enhance model transparency, which is essential for trust and compliance in healthcare systems. These extensions would strengthen the applicability of the proposed system in large-scale, real-world IoMT deployments.

REFERENCES

CHAPTER 10

REFERENCES

1. D. V. Dimitrov, “Medical Internet of Things and big data in healthcare,” *Healthcare Informat. Res.*, vol. 22, no. 3, p. 156, 2021, doi: 10.4258/hir.2016.22.3.156.
2. P. Manickam, S. A. Mariappan, S. M. Murugesan, S. Hansda, A. Kaushik, R. Shinde, and S. P. Thipperudraswamy, “Artificial intelligence (AI) and Internet of Medical Things (IoMT) assisted biomedical systems for intelligent healthcare,” *Biosensors*, vol. 12, no. 8, p. 562, Jul. 2022, doi: 10.3390/bios12080562.
3. W. Sun, Z. Cai, Y. Li, F. Liu, S. Fang, and G. Wang, “Security and privacy in the medical Internet of Things: A review,” *Secur. Commun. Netw.*, vol. 2018, pp. 1–9, Jul. 2018, doi: 10.1155/2018/5978636.
4. T. Yaqoob, H. Abbas, and M. Atiquzzaman, “Security vulnerabilities, attacks, countermeasures, and regulations of networked medical devices—A review,” *IEEE Commun. Surveys Tuts.*, vol. 21, no. 4, pp. 3723–3768, 4th Quart., 2019, doi: 10.1109/COMST.2019.2914094.
5. T. Levy-Loboda, E. Sheetrit, I. F. Liberty, A. Haim, and N. Nissim, “Personalized insulin dose manipulation attack and its detection using interval-based temporal patterns and machine learning algorithms,” *J. Biomed. Informat.*, vol. 132, Aug. 2022, Art. no. 104129, doi: 10.1016/j.jbi.2022.104129.
6. P. Sharma, S. Jain, S. Gupta, and V. Chamola, “Role of machine learning and deep learning in securing 5G-driven industrial IoT applications,” *Ad Hoc Netw.*, vol. 123, Dec. 2021, Art. no. 102685, doi: 10.1016/j.adhoc.2021.102685.
7. N. Moustafa and J. Slay, “UNSW-NB15: A comprehensive data set for network intrusion detection systems (UNSW-NB15 network data set),” in *Proc. Mil. Commun. Inf. Syst. Conf. (MilCIS)*, Nov. 2015, pp. 1–6. [Online]. Available: <https://ieeexplore.ieee.org/document/7348942>
8. E. Šabić, D. Keeley, B. Henderson, and S. Nannemann, “Healthcare and anomaly detection: Using machine learning to predict anomalies in heart rate data,” *AI Soc.*, vol. 36, no. 1, pp. 149–158, May 2020, doi: 10.1007/s00146-020-00985-1.

9. S. Park, K. H. Lee, B. Ko, and N. Kim, “Unsupervised anomaly detection with generative adversarial networks in mammography,” *Sci. Rep.*, vol. 13, no. 1, pp. 1–10, Feb. 2023, doi: 10.1038/s41598-023-29521-z.
10. M. Kavitha, P. V. V. S. Srinivas, P. S. L. Kalyampudi, S. F. Choragudi, and S. Srinivasulu, “Machine learning techniques for anomaly detection in smart healthcare,” in *Proc. 3rd Int. Conf. Inventive Res. Comput. Appl. (ICIRCA)*, Sep. 2021, pp. 1350–1356.
11. L. Fang, Y. Li, Z. Liu, C. Yin, M. Li, and Z. J. Cao, “A practical model based on anomaly detection for protecting medical IoT control services against external attacks,” *IEEE Trans. Ind. Informat.*, vol. 17, no. 6, pp. 4260–4269, Jun. 2021, doi: 10.1109/TII.2020.3011444.
12. T. Alsolami, B. Alsharif, and M. Ilyas, “Enhancing cybersecurity in healthcare: Evaluating ensemble learning models for intrusion detection in the Internet of Medical Things,” *Sensors*, vol. 24, no. 18, p. 5937, Sep. 2024, doi: 10.3390/s24185937.
13. A. A. Hady, A. Ghubaish, T. Salman, D. Unal, and R. Jain, “Intrusion detection system for healthcare systems using medical and network data: A comparison study,” *IEEE Access*, vol. 8, pp. 106576–106584, 2020, doi: 10.1109/ACCESS.2020.3000421.
14. I. Ullah and Q. H. Mahmoud, “Design and development of a deep learning-based model for anomaly detection in IoT networks,” *IEEE Access*, vol. 9, pp. 103906–103926, 2021, doi: 10.1109/ACCESS.2021.3094024.
15. S. Das, M. Ashrafuzzaman, F. T. Sheldon, and S. Shiva, “Ensembling supervised and unsupervised machine learning algorithms for detecting distributed denial of service attacks,” *Algorithms*, vol. 17, no. 3, p. 99, Feb. 2024, doi: 10.3390/a17030099.
16. Y. Gu, K. Li, Z. Guo, and Y. Wang, “Semi-supervised K-means DDoS detection method using hybrid feature selection algorithm,” *IEEE Access*, vol. 7, pp. 64351–64365, 2019, doi: 10.1109/ACCESS.2019.2917532.
17. Y. Meidan, M. Bohadana, Y. Mathov, Y. Mirsky, A. Shabtai, D. Breitenbacher, and Y. Elovici, “N-BaIoT—Network-Based detection of IoT botnet attacks using deep autoencoders,” *IEEE Pervasive Comput.*, vol. 17, no. 3, pp. 12–22, Jul. 2018, doi: 10.1109/MPRV.2018.03367731.

18. N. Ravi and S. M. Shalinie, "Learning-driven detection and mitigation of DDoS attack in IoT via SDN-cloud architecture," *IEEE Internet Things J.*, vol. 7, no. 4, pp. 3559–3570, Apr. 2020, doi: 10.1109/JIOT.2020.2973176.
19. R. Doshi, N. Aphorpe, and N. Feamster, "Machine learning DDoS detection for consumer Internet of Things devices," in *Proc. IEEE Secur. Privacy Workshops (SPW)*, May 2018, pp. 29–35.
20. Z. K. Maseer, R. Yusof, N. Bahaman, S. A. Mostafa, and C. F. M. Foozy, "Benchmarking of machine learning for anomaly based intrusion detection systems in the CICIDS2017 dataset," *IEEE Access*, vol. 9, pp. 22351–22370, 2021, doi: 10.1109/ACCESS.2021.3056614.
21. K. Choi, J. Yi, C. Park, and S. Yoon, "Deep learning for anomaly detection in time-series data: Review, analysis, and guidelines," *IEEE Access*, vol. 9, pp. 120043–120065, 2021, doi: 10.1109/ACCESS.2021.3107975.
22. B. Luo, H. Wang, H. Liu, B. Li, and F. Peng, "Early fault detection of machine tools based on deep learning and dynamic identification," *IEEE Trans. Ind. Electron.*, vol. 66, no. 1, pp. 509–518, Jan. 2019, doi: 10.1109/TIE.2018.2807414.
23. G. Abdelmoumin, D. B. Rawat, and A. Rahman, "On the performance of machine learning models for anomaly-based intelligent intrusion detection systems for the Internet of Things," *IEEE Internet Things J.*, vol. 9, no. 6, pp. 4280–4290, Mar. 2022, doi: 10.1109/JIOT.2021.3103829.
24. I. G. A. Poornima and B. Paramasivan, "Anomaly detection in wireless sensor network using machine learning algorithm," *Comput. Commun.*, vol. 151, pp. 331–337, Feb. 2020, doi: 10.1016/j.comcom.2020.01.005.
25. J. Alsamiri and K. Alsubhi, "Internet of Things cyber attacks detection using machine learning," *Int. J. Adv. Comput. Sci. Appl.*, vol. 10, no. 12, 2019, doi: 10.14569/ijacsa.2019.0101280.
26. M. Hasan, M. S. Rahman, H. Janicke, and I. H. Sarker, "Detecting anomalies in blockchain transactions using machine learning classifiers and explainability analysis," *Blockchain, Res. Appl.*, vol. 5, no. 3, Sep. 2024, Art. no. 100207, doi: 10.1016/j.bcra.2024.100207.

27. T. Cover and P. Hart, "Nearest neighbor pattern classification," *IEEE Trans. Inf. Theory*, vol. IT-13, no. 1, pp. 21–27, Jan. 1967, doi: 10.1109/TIT.1967.1053964.
28. S. A. Mulay, P. R. Devale, and G. V. Garje, "Intrusion detection system using support vector machine and decision tree," *Int. J. Comput. Appl.*, vol. 3, no. 3, pp. 40–43, Jun. 2010, doi: 10.5120/758-993.
29. G. J. M. Rosa, "The elements of statistical learning: Data mining, inference, and prediction by HASTIE, T., TIBSHIRANI, R., and FRIEDMAN, J," *Biometrics*, vol. 66, no. 4, p. 1315, Dec. 2010, doi: 10.1111/j.1541-0420.2010.01516.x.
30. N. Farnaaz and M. A. Jabbar, "Random forest modeling for network intrusion detection system," *Proc. Comput. Sci.*, vol. 89, pp. 213–217, Jun. 2016.